Notes

George

October 30, 2026

Theorem 1. *Let X be a scheme, let $\mathcal{U} = \{U_\lambda\}_{\lambda\in\Lambda}$ be an open cover of X, and fix a rank n. Then there is a bijection*

$$\left\{\begin{matrix}\text{locally free } \mathcal{O}_X\text{-modules of rank } n \\ \text{trivialised over } \mathcal{U}\end{matrix}\right\} / \cong \quad \longleftrightarrow \quad \check{H}^1(\mathcal{U}, \mathrm{GL}_n(\mathcal{O}_X)). \tag{1}$$

Proof. Write $U_{\lambda\mu} = U_\lambda \cap U_\mu$ and $U_{\lambda\mu\nu} = U_\lambda \cap U_\mu \cap U_\nu$. We use $\mathcal{O}_U = \mathcal{O}_X|_U$ for each open subset $U \subseteq X$. To specify the distinction between matrices and sheaf morphisms, if $A \in \mathrm{GL}_n(\mathcal{O}_X(U))$, let $\mathrm{m}_A : \mathcal{O}_U^{\oplus n} \longrightarrow \mathcal{O}_U^{\oplus n}$ denote the automorphism of $\mathcal{O}_U$-modules whose component on an open set $V \subseteq U$ is $(\mathrm{m}_A)_V(s) = (A|_V)s$. Note that $\mathrm{Aut}_{\mathcal{O}_U}(\mathcal{O}_U^{\oplus n}) \cong \mathrm{GL}_n(\mathcal{O}_X(U)) = \mathrm{GL}_n(\mathcal{O}_X)(U)$, where $\mathrm{Aut}_{\mathcal{O}_U}$ denotes automorphisms of sheaves of $\mathcal{O}_U$-modules.

We first construct the forward map α. Let $\mathscr{F}$ be a locally free $\mathcal{O}_X$-module of rank n trivialised over $\mathcal{U}$. Choose isomorphisms of $\mathcal{O}_{U_\lambda}$-modules

$$\phi_\lambda : \mathcal{O}_{U_\lambda}^{\oplus n} \longrightarrow \mathscr{F}|_{U_\lambda}.$$

For every open set $V \subseteq U_\lambda$, the corresponding map on sections is an isomorphism of $\mathcal{O}_X(V)$-modules

$$(\phi_\lambda)_V : \mathcal{O}_X(V)^{\oplus n} \longrightarrow \mathscr{F}(V).$$

Define

$$T_{\lambda\mu} \in \mathrm{GL}_n(\mathcal{O}_X(U_{\lambda\mu}))$$

by the identity of sheaf morphisms

$$\mathrm{m}_{T_{\lambda\mu}} = (\phi_\lambda|_{U_{\lambda\mu}})^{-1} \circ (\phi_\mu|_{U_{\lambda\mu}}).$$

On $U_{\lambda\mu\nu}$, we have

$$\mathrm{m}_{T_{\lambda\mu}T_{\mu\nu}} = (\phi_\lambda^{-1} \circ \phi_\mu) \circ (\phi_\mu^{-1} \circ \phi_\nu) = \phi_\lambda^{-1} \circ \phi_\nu = \mathrm{m}_{T_{\lambda\nu}}.$$

Hence

$$T_{\lambda\mu}T_{\mu\nu} = T_{\lambda\nu}.$$

We also have $T_{\lambda\lambda} = I_n$ and $T_{\mu\lambda} = T_{\lambda\mu}^{-1}$. Therefore

$$(T_{\lambda\mu})_{\lambda,\mu\in\Lambda} \in \check{Z}^1(\mathcal{U}, \mathrm{GL}_n(\mathcal{O}_X)).$$

Now suppose that $\mathscr{F}'$ is another locally free $\mathcal{O}_X$-module of rank n trivialised over $\mathcal{U}$, with chosen trivialisations ϕ'_λ and transition matrices $T'_{\lambda\mu}$, and let

$$\psi : \mathscr{F} \longrightarrow \mathscr{F}'$$

be an isomorphism of $\mathcal{O}_X$-modules. Write $\psi_\lambda = \psi|_{U_\lambda}$ and define $h_\lambda \in \mathrm{GL}_n(\mathcal{O}_X(U_\lambda))$ by

$$\mathrm{m}_{h_\lambda} = (\phi'_\lambda)^{-1} \circ \psi_\lambda \circ \phi_\lambda.$$

Thus

$$(h_\lambda)_{\lambda\in\Lambda} \in \check{C}^0(\mathcal{U}, \mathrm{GL}_n(\mathcal{O}_X)) = \prod_{\lambda\in\Lambda} \mathrm{GL}_n(\mathcal{O}_X(U_\lambda)).$$

On $U_{\lambda\mu}$, the restrictions of ψ_λ and ψ_μ agree, so

$$\begin{aligned} \mathrm{m}_{h_\lambda T_{\lambda\mu}} &= (\phi'_\lambda)^{-1} \circ \psi_\lambda \circ \phi_\mu \\ &= (\phi'_\lambda)^{-1} \circ \psi_\mu \circ \phi_\mu \\ &= \mathrm{m}_{T'_{\lambda\mu} h_\mu}. \end{aligned}$$

Consequently, $h_\lambda T_{\lambda\mu} = T'_{\lambda\mu} h_\mu$, which is precisely the equivalence relation defining $\check{H}^1(\mathcal{U}, \mathrm{GL}_n(\mathcal{O}_X))$. We therefore obtain a well-defined map

$$\alpha : [\mathscr{F}] \longmapsto [(T_{\lambda\mu})].$$

To prove injectivity, suppose that the cocycles associated to $\mathscr{F}$ and $\mathscr{F}'$ represent the same class. Thus there exists a 0-cochain $(h_\lambda)_{\lambda\in\Lambda}$ such that

$$h_\lambda T_{\lambda\mu} = T'_{\lambda\mu} h_\mu$$

on every overlap. Define isomorphisms of $\mathcal{O}_{U_\lambda}$-modules

$$\psi_\lambda = \phi'_\lambda \circ \mathrm{m}_{h_\lambda} \circ \phi_\lambda^{-1} : \mathscr{F}|_{U_\lambda} \longrightarrow \mathscr{F}'|_{U_\lambda}.$$

On $U_{\lambda\mu}$, we have

$$\begin{aligned} \psi_\lambda &= \phi'_\lambda \circ \mathrm{m}_{h_\lambda} \circ \mathrm{m}_{T_{\lambda\mu}} \circ \phi_\mu^{-1} \\ &= \phi'_\lambda \circ \mathrm{m}_{T'_{\lambda\mu}} \circ \mathrm{m}_{h_\mu} \circ \phi_\mu^{-1} \\ &= \phi'_\mu \circ \mathrm{m}_{h_\mu} \circ \phi_\mu^{-1} \\ &= \psi_\mu. \end{aligned}$$

Hence the ψ_λ glue to an isomorphism $\psi : \mathscr{F} \to \mathscr{F}'$ of $\mathcal{O}_X$-modules; their inverses also agree on overlaps and glue to its inverse. Thus α is injective.

For the reverse construction, let

$$(T_{\lambda\mu})_{\lambda,\mu\in\Lambda} \in \check{Z}^1(\mathcal{U}, \mathrm{GL}_n(\mathcal{O}_X)).$$

For each open set $V \subseteq X$, define

$$\mathscr{F}(V) = \left\{ (s_\lambda)_{\lambda\in\Lambda} \in \prod_{\lambda\in\Lambda} \mathcal{O}_X(V \cap U_\lambda)^{\oplus n} \;\middle|\; s_\lambda = T_{\lambda\mu} s_\mu \text{ on } V \cap U_{\lambda\mu} \right\}. \tag{2}$$

For open sets $W \subseteq V$, define restriction maps

$$\mathscr{F}(V) \longrightarrow \mathscr{F}(W), \qquad (s_\lambda)_\lambda \longmapsto (s_\lambda|_{W\cap U_\lambda})_\lambda.$$

These maps preserve the compatibility condition and satisfy the presheaf axioms. Addition is defined componentwise, and for $f \in \mathcal{O}_X(V)$ scalar multiplication is defined by

$$f(s_\lambda)_\lambda = \big((f|_{V\cap U_\lambda})s_\lambda\big)_\lambda.$$

These operations preserve the compatibility condition because multiplication by each $T_{\lambda\mu}$ is linear. Thus $\mathscr{F}$ is a presheaf of $\mathcal{O}_X$-modules. The separation and gluing axioms follow componentwise from those for $\mathcal{O}_X^{\oplus n}$.

We now construct isomorphisms of sheaves

$$\phi_\lambda : \mathcal{O}_{U_\lambda}^{\oplus n} \longrightarrow \mathscr{F}|_{U_\lambda}.$$

For every open set $V \subseteq U_\lambda$, define

$$(\phi_\lambda)_V : \mathcal{O}_X(V)^{\oplus n} \longrightarrow \mathscr{F}(V), \qquad s \longmapsto \Big(T_{\mu\lambda}|_{V\cap U_\mu}\, s|_{V\cap U_\mu}\Big)_{\mu\in\Lambda}.$$

This family satisfies the compatibility condition because

$$T_{\mu\nu}T_{\nu\lambda} = T_{\mu\lambda}.$$

The map $(\phi_\lambda)_V$ is an isomorphism of $\mathcal{O}_X(V)$-modules, with inverse

$$(s_\mu)_{\mu\in\Lambda} \longmapsto s_\lambda.$$

Indeed, $T_{\lambda\lambda} = I_n$, and every compatible family satisfies $s_\mu = T_{\mu\lambda}s_\lambda$. These maps commute with restriction, so ϕ_λ is an isomorphism of sheaves of $\mathcal{O}_{U_\lambda}$-modules. Therefore $\mathscr{F}$ is locally free of rank n and is trivialised over $\mathcal{U}$.

On $U_{\lambda\mu}$, the λ-component of $\phi_\mu(s)$ is $T_{\lambda\mu}s$. Consequently,

$$\phi_\lambda^{-1} \circ \phi_\mu = \mathrm{m}_{T_{\lambda\mu}},$$

so the transition matrices of the constructed sheaf are exactly the original $T_{\lambda\mu}$.

Finally, if

$$T'_{\lambda\mu} = h_\lambda T_{\lambda\mu} h_\mu^{-1},$$

the componentwise maps

$$\mathscr{F}(V) \longrightarrow \mathscr{F}'(V), \qquad (s_\lambda)_\lambda \longmapsto \big((h_\lambda|_{V\cap U_\lambda})s_\lambda\big)_\lambda$$

define an isomorphism of $\mathcal{O}_X$-modules, with inverse given componentwise by h_λ^{-1}. Thus the reverse construction defines a well-defined map

$$\beta : [(T_{\lambda\mu})] \longmapsto [\mathscr{F}].$$

Since $\alpha \circ \beta = \mathrm{Id}$, the map α is surjective. Together with injectivity, this proves the claimed bijection. $\square$

Theorem 2. *If X is a topological space then there is a bijection*

$$\left\{\begin{matrix}\text{real vector}\\ \text{bundles on } X\end{matrix}\right\}/\cong \quad\longleftrightarrow\quad \left\{\begin{matrix}\text{locally free sheaves}\\ \text{of } \mathcal{C}_{X,\mathbb{R}}\text{-modules on } X\end{matrix}\right\}/\cong \tag{3}$$

Proof. We first construct the $\to$ map α. Let $E \xrightarrow{\pi} X$ be a real vector bundle of rank n, trivialised over an open cover $\mathcal{U} = \{U_\lambda\}_{\lambda\in\Lambda}$ by homeomorphisms

$$h_\lambda : \pi^{-1}(U_\lambda) \longrightarrow U_\lambda \times \mathbb{R}^n.$$

For each $x \in X$, write $E_x = \pi^{-1}(x)$. The vector space structures on the fibres define maps

$$a : E \times_X E \longrightarrow E, \qquad (e, e') \longmapsto e + e', \tag{4}$$

$$m : \mathbb{R} \times E \longrightarrow E, \qquad (r, e) \longmapsto re, \tag{5}$$

where

$$E \times_X E = \{(e, e') \in E \times E \mid \pi(e) = \pi(e')\}$$

has the subspace topology. These maps are globally well defined: addition is performed in the common fibre of e and e', and scalar multiplication is performed in the fibre containing e. In particular, their definitions do not depend on any choice of trivialisation.

We claim that both maps are continuous. For each λ, the trivialisation h_λ induces a homeomorphism

$$\pi^{-1}(U_\lambda) \times_{U_\lambda} \pi^{-1}(U_\lambda) \longrightarrow U_\lambda \times \mathbb{R}^n \times \mathbb{R}^n,$$

sending (e, e') to (x, v, w) when $h_\lambda(e) = (x, v)$ and $h_\lambda(e') = (x, w)$. Since h_λ is linear on each fibre, the restriction of a is represented in these coordinates by

$$(x, v, w) \longmapsto (x, v + w).$$

Likewise, using the homeomorphism

$$\mathbb{R} \times \pi^{-1}(U_\lambda) \longrightarrow U_\lambda \times \mathbb{R} \times \mathbb{R}^n, \qquad (r, e) \longmapsto (\pi(e), r, \mathrm{pr}_2(h_\lambda(e))),$$

the restriction of m is represented by

$$(x, r, v) \longmapsto (x, rv).$$

These coordinate maps are continuous. Moreover, the sets $\pi^{-1}(U_\lambda) \times_{U_\lambda} \pi^{-1}(U_\lambda)$ form an open cover of $E \times_X E$, and the sets $\mathbb{R} \times \pi^{-1}(U_\lambda)$ form an open cover of $\mathbb{R} \times E$. Thus a and m are continuous on their entire domains.

For each open set $V \subseteq X$, define

$$\mathcal{E}(V) = \{s : V \to E \text{ continuous} \mid \pi \circ s = i_{V,X}\},$$

with restriction maps given by function restriction. Define addition and multiplication by $f \in \mathcal{C}_{X,\mathbb{R}}(V)$ pointwise:

$$(s + t)(x) = s(x) + t(x), \qquad (fs)(x) = f(x)s(x).$$

These operations produce sections over V, since their values lie in E_x, which implies that $\pi \circ s = i_{V,X}$ can be verified locally. They also preserve continuity: the map

$$V \longrightarrow E \times_X E, \qquad x \longmapsto (s(x), t(x)),$$

is continuous, and its composition with a is $s + t$; similarly, fs is the composition of the continuous map

$$V \longrightarrow \mathbb{R} \times E, \qquad x \longmapsto (f(x), s(x)),$$

with m.

The zero section $x \mapsto 0_{E_x}$ is continuous, since in every trivialisation it is given by $x \mapsto (x, 0)$, and the additive inverse of s is the continuous section $(-1)s$. All module axioms hold pointwise by the vector space axioms in each fibre. Hence $\mathcal{E}(V)$ is a $\mathcal{C}_{X,\mathbb{R}}(V)$-module. These operations commute with restriction, so $\mathcal{E}$ is a presheaf of $\mathcal{C}_{X,\mathbb{R}}$-modules. Continuous sections that agree on overlaps glue uniquely to a continuous section, so $\mathcal{E}$ is a sheaf of $\mathcal{C}_{X,\mathbb{R}}$-modules.

For each λ, the trivialisation h_λ induces an isomorphism of sheaves of $\mathcal{C}_{X,\mathbb{R}}|_{U_\lambda}$-modules

$$\phi_\lambda : \mathcal{C}_{X,\mathbb{R}}^{\oplus n}|_{U_\lambda} \longrightarrow \mathcal{E}|_{U_\lambda}.$$

For every open set $V \subseteq U_\lambda$, define $(\phi_\lambda)_V$ by sending a continuous map $f : V \to \mathbb{R}^n$ to the continuous section

$$x \longmapsto h_\lambda^{-1}(x, f(x)).$$

It has an inverse, given by

$$(\phi_\lambda)_V^{-1}(s) = \big(x \longmapsto \mathrm{pr}_2(h_\lambda(s(x)))\big).$$

Both constructions commute with restrictions, so ϕ_λ is an isomorphism, and $\mathcal{E}$ is locally free of rank n.

Finally, if $F : E \to \widetilde{E}$ is a vector bundle isomorphism over X, composition with F induces an isomorphism of sheaves of $\mathcal{C}_{X,\mathbb{R}}$-modules

$$\mathcal{E} \longrightarrow \widetilde{\mathcal{E}}, \qquad s \longmapsto F \circ s.$$

Consequently, the assignment $\alpha([E]) = [\mathcal{E}]$ is well defined on isomorphism classes.

We now consider the $\leftarrow$ direction, β. Let $\mathcal{E}$ be a locally free sheaf of $\mathcal{C}_{X,\mathbb{R}}$-modules, trivialised over an open cover $\mathcal{U} = \{U_\lambda\}_{\lambda \in \Lambda}$ by isomorphisms of sheaves of $\mathcal{C}_{X,\mathbb{R}}$-modules

$$\phi_\lambda : \mathcal{C}_{X,\mathbb{R}}^{\oplus n}|_{U_\lambda} \longrightarrow \mathcal{E}|_{U_\lambda}. \tag{6}$$

On $U_{\lambda\mu}$, we have a matrix

$$T_{\lambda\mu} \in \mathrm{GL}_n(\mathbb{C}_{X,\mathbb{R}}(U_{\lambda\mu})), \tag{7}$$

corresponding to the isomorphism of sheaves $\mathrm{m}_{T_{\lambda\mu}} = \phi_\lambda^{-1}\phi_\mu$. Composition of these morphisms gives

$$T_{\lambda\lambda} = I_n, \qquad T_{\mu\lambda} = T_{\lambda\mu}^{-1}, \qquad T_{\lambda\mu}T_{\mu\nu} = T_{\lambda\nu}, \tag{8}$$

Let us define the total space

$$E = \left(\coprod_{\lambda \in \Lambda} U_\lambda \times \mathbb{R}^n \right) \Big/ \sim,$$

where

$$(x, v)_\lambda \sim (y, w)_\mu \quad \Longleftrightarrow \quad x = y \quad \text{and} \quad v = T_{\lambda\mu}(x)w.$$

By Equation 8, this relation is reflexive, symmetric, and transitive. Let us denote the quotient map by

$$q : \coprod_{\lambda \in \Lambda} U_\lambda \times \mathbb{R}^n \longrightarrow E$$

and define

$$\pi : E \longrightarrow X, \qquad [(x, v)_\lambda] \longmapsto x.$$

Gluing along the transition homeomorphisms gives local trivialisations

$$h_\lambda : \pi^{-1}(U_\lambda) \longrightarrow U_\lambda \times \mathbb{R}^n, \qquad [(x, v)_\mu] \longmapsto (x, T_{\lambda\mu}(x)v),$$

with inverses

$$(x, v) \longmapsto [(x, v)_\lambda].$$

The vector space structures on $\mathbb{R}^n$ induce vector space structures on the fibres of π, independently of the chosen chart, because every $T_{\lambda\mu}(x)$ is linear. Thus $E \xrightarrow{\pi} X$ is a real vector bundle of rank n.

Let $\widetilde{\mathcal{E}}$ be its sheaf of continuous sections. For an open set $V \subseteq X$ and $s \in \mathcal{E}(V)$, write

$$\overline{s}_\lambda = (\phi_\lambda^{-1})_{V \cap U_\lambda}(s|_{V \cap U_\lambda}).$$

These coordinate sections satisfy

$$\overline{s}_\lambda = T_{\lambda\mu}\overline{s}_\mu$$

on overlaps, so the formula

$$\eta_V(s)(x) = [(x, \overline{s}_\lambda(x))_\lambda], \qquad x \in V \cap U_\lambda,$$

defines a continuous section of E over V.

Conversely, a continuous section t of E over V has coordinate maps

$$\overline{t}_\lambda(x) = \mathrm{pr}_2(h_\lambda(t(x))), \qquad x \in V \cap U_\lambda,$$

satisfying the same compatibility condition. Hence

$$(\phi_\lambda)_{V \cap U_\lambda}(\overline{t}_\lambda) \in \mathcal{E}(V \cap U_\lambda)$$

agree on overlaps and glue uniquely to an element of $\mathcal{E}(V)$. This construction is inverse to η_V. These maps are $\mathcal{C}_{X,\mathbb{R}}(V)$-linear and commute with restriction, giving an isomorphism

$$\eta : \mathcal{E} \xrightarrow{\sim} \widetilde{\mathcal{E}}.$$

To check independence of choices, suppose that $\theta : \mathcal{E} \to \mathcal{E}'$ is an isomorphism, and choose local trivialisations ϕ_λ and ϕ'_λ on a common refinement of their covers. Let H_λ be the matrix representing the sheaf isomorphism

$$(\phi'_\lambda)^{-1} \circ \theta|_{U_\lambda} \circ \phi_\lambda.$$

Then

$$H_\lambda T_{\lambda\mu} = T'_{\lambda\mu} H_\mu,$$

so the local maps

$$(x, v) \longmapsto (x, H_\lambda(x)v)$$

glue to a vector bundle isomorphism between the reconstructed bundles. Taking $\theta = \mathrm{Id}_\mathcal{E}$ also proves independence of the chosen trivialisations; passing to a refinement does not change the glued bundle up to isomorphism. Therefore

$$\beta : [\mathcal{E}] \longmapsto [E]$$

is well defined, and the isomorphism η proves that $\alpha \circ \beta = \mathrm{Id}$.

Finally, if we start with a vector bundle E with trivialisations h_λ, the induced trivialisations of its section sheaf give the same transition matrices. The reconstructed bundle is therefore isomorphic to E via

$$[(x, v)_\lambda] \longmapsto h_\lambda^{-1}(x, v).$$

Thus $\beta \circ \alpha = \mathrm{Id}$ on isomorphism classes, completing the proof.

□